UPSA
UNIVERSITY OF PROFESSIONAL STUDIES, ACCRA
Scholarship with Professionalism

NETWORK AND SYSTEMS ADMINISTRATION

WEEK 1: INTRODUCTION TO NETWORK AND SYSTEMS ADMINISTRATION

Roles and Ethics • Privileged Access • Kali Linux Lab Orientation • Basic Administrative Tools • Git • Responsible AI

Practical Environment: Kali Linux

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WEEK 1 ROADMAP

From organisational need to administrative evidence

1. **Organisation** — Users, goals, services
2. **Infrastructure** — Systems + network
3. **Administration** — Configure, monitor, protect
4. **Control** — Document + manage change
5. **Evidence** — Inventory + diagram

WHAT YOU WILL LEARN

Meaning and scope • administrator roles • infrastructure components • documentation • change control • network–systems relationship

WHAT YOU WILL DO

Run six Linux inventory commands • interpret evidence • form 6–7 groups • model a small Ghanaian organisation • present an infrastructure diagram

WEEK 1 LEARNING OUTCOMES

By the end of the three-hour session, students should be able to:

- Explain network and systems administration in organisational terms.
- Distinguish major administrator roles while recognising their overlap.
- Identify users, systems, network resources and services in a small IT environment.
- Explain why documentation and change control are essential administrative practices.
- Describe how network work and systems work depend on each other.
- Use basic Linux commands to record a system and network inventory.
- Translate a Ghanaian organisational scenario into a simple infrastructure diagram.

WHAT DOES “ADMINISTRATION” MEAN?

Administration is the disciplined management of IT resources so that services remain useful, reliable, secure and supportable.

CONFIGURE

Set up systems, interfaces, services, accounts and policies so the environment can perform its intended work.

OPERATE

Keep services available, observe resource use, respond to routine needs and maintain the environment.

CONTROL

Protect access, document decisions, manage changes, preserve evidence and reduce avoidable risk.

Administration connects technology to organisational service delivery.

SYSTEMS ADMINISTRATION

Focus: computing hosts, operating systems, resources, services and users.

- Servers and workstations: installation, configuration and lifecycle.
- User accounts, groups, permissions and privileged access.
- CPU, memory, storage, file systems, processes and services.
- Software installation, updates, patches and local configuration.
- Backups, restoration readiness, monitoring and troubleshooting.
- Security controls, logs, documentation and operational procedures.

ADMINISTRATOR'S QUESTION

"Is this host healthy, secure, correctly configured and able to deliver its service?"

EVIDENCE

Configuration • logs • process/service status • resource use • account records • backup records

NETWORK ADMINISTRATION

Focus: connectivity, addressing, network devices, traffic paths and network services.

- Interfaces, links and network devices such as switches, routers and access points.
- IP configuration, routes, segmentation and connectivity between hosts.
- Network services such as DNS and DHCP that support communication.
- Availability, fault detection, performance observation and capacity planning.
- Secure network access, filtering and controlled administrative access.
- Network documentation, diagrams, configuration records and troubleshooting evidence.

ADMINISTRATOR'S QUESTION

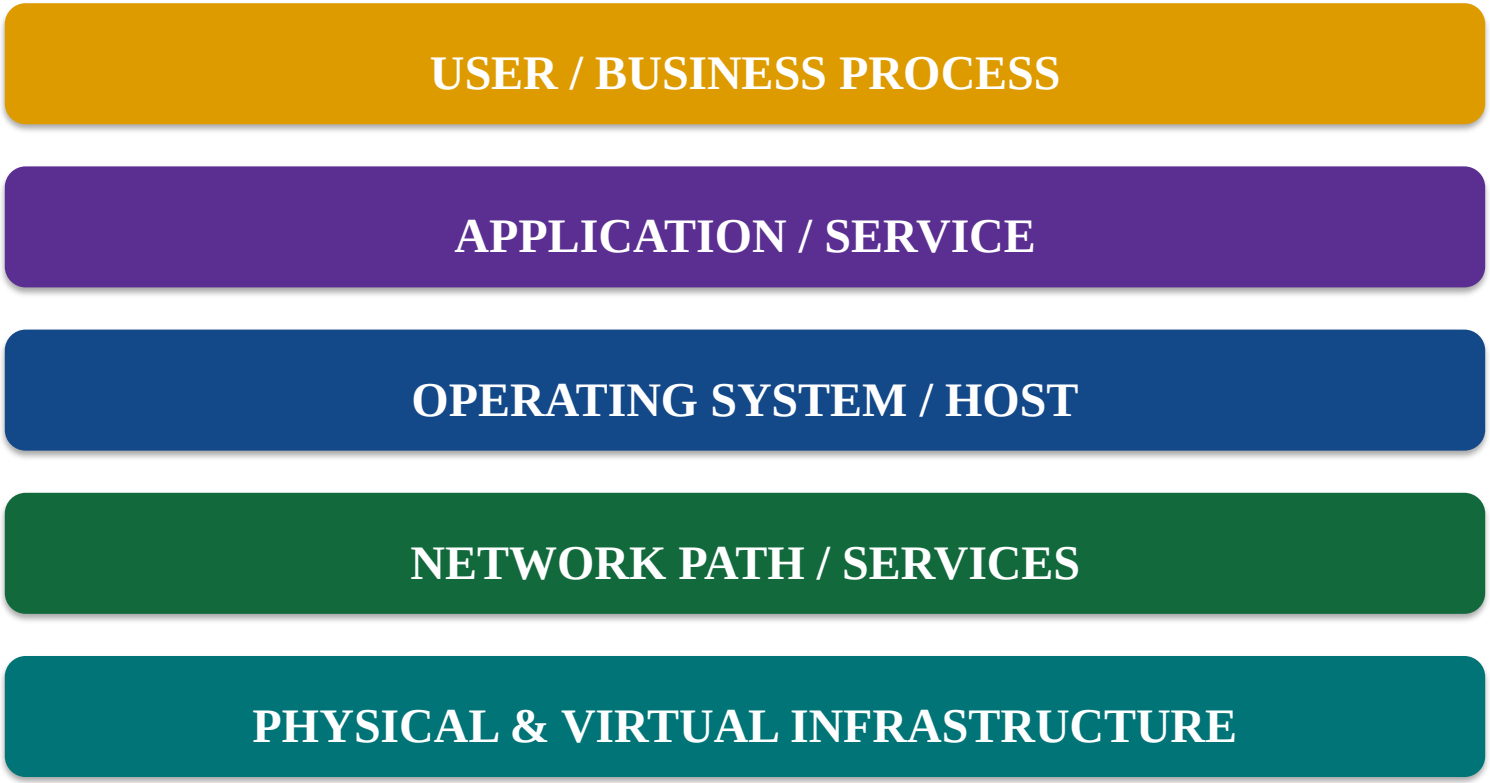
"Can authorised users and systems communicate reliably through the intended path?"

EVIDENCE

Link state • addresses • routes • device configs • logs • diagrams • test results

NETWORK + SYSTEMS: ONE SERVICE ENVIRONMENT

A user experiences a service, not departmental boundaries.



A fault at any layer can make the same service "unavailable" to the user.

THE ADMINISTRATOR'S INFRASTRUCTURE MAP

Think in categories before thinking in commands.

USERS

Employees, students, customers, administrators, service accounts

SYSTEMS

Servers, workstations, VMs, operating systems, storage

NETWORK

Interfaces, switches, routers, APs, links, addresses, routes

SERVICES

Web, file, database, DNS, DHCP, email, authentication

CONTROLS

Accounts, permissions, firewall rules, policies, backups

EVIDENCE

Inventories, diagrams, logs, tickets, change records, configs

ADMINISTRATOR ROLES

Roles may be separate in a large enterprise and combined in a small organisation.

Systems Administrator

Hosts, OS, users, services, storage, updates

Network Administrator

Connectivity, devices, IPs, routes, network services

Security Administrator

Controls, monitoring, hardening, audit evidence

Cloud / Virtualisation Administrator

VMs, virtual networks, shared resources

Support / Service Desk

User incidents, first-line diagnosis, escalation

IT Operations / Infrastructure Team

Availability, maintenance, monitoring, coordinated response

CORE DUTIES OF A SYSTEM ADMINISTRATOR

Chapter 1 of the handbook provides a useful operational checklist.

- Provision and retire user accounts responsibly.
- Add, remove and support hardware.
- Perform backups and ensure restoration is possible.
- Install, test, upgrade and patch software.
- Monitor systems and resource availability.
- Troubleshoot failures and escalate when necessary.
- Maintain accurate local documentation.
- Monitor security and enforce policy.

KEY IDEA

The job is not complete because a task was automated or delegated. The administrator remains responsible for verifying

OPERATIONS

Stable service requires routine care, evidence, controlled changes and readiness for failure.

PROFESSIONAL RESPONSIBILITY

Administrative power must be matched by accountability.

AUTHORITY

Administrators may access systems, configurations, logs and sensitive resources that ordinary users cannot.

DUTY OF CARE

Use access only for authorised work. Protect credentials, data and service availability.

ACCOUNTABILITY

Actions should be explainable and, where appropriate, recorded so that changes can be traced.

A technically successful action can still be professionally wrong if it is unauthorised, undocumented or unnecessarily risky.

DOCUMENTATION: THE ORGANISATION'S MEMORY

If knowledge exists only in one administrator's head, it is an operational risk.

INVENTORY

What exists? Hosts, devices, interfaces, services, owners.

CONFIGURATION

How is it set up? Key settings, dependencies and locations.

NETWORK DIAGRAM

How are systems connected? Paths, segments and critical links.

PROCEDURES

How do we perform routine and emergency tasks consistently?

MAINTENANCE RECORD

What was changed, repaired, patched or replaced?

BACKUP / RECOVERY RECORD

What is protected, when, where and has restoration been tested?

CHANGE CONTROL

A change should be intentional, authorised, testable and reversible.

1. REQUEST

What needs to change and why?

2. ASSESS

Impact, risk, dependencies, timing.

3. AUTHORISE

Who approves the change?

4. IMPLEMENT

Make the planned change carefully.

5. VERIFY

Did it produce the intended result?

6. RECORD /

Document success or restore prior state.

Operational principle: make one controlled change at a time, test it, and document what happened.

CHANGE CONTROL EXAMPLE

Scenario: the organisation must change the IP address of an internal application server.

- Request: application team needs the server moved to a new subnet.
- Assess dependencies: DNS record, firewall rules, routes, monitoring, client configuration and documentation.
- Authorise: confirm business owner and maintenance window.
- Implement: change the address using the approved procedure.
- Verify: interface state, route, DNS resolution, service reachability and user test.
- Record: update inventory and network diagram; retain evidence.
- Rollback: restore the previous address if verification fails.

PRACTICAL ENVIRONMENT: LINUX VM

Week 1 begins with observation, not modification.

HOST

The Linux virtual machine has a hostname, kernel, operating system and hardware/virtual hardware characteristics.

NETWORK

The VM exposes network interfaces, addresses and routes that depend partly on the virtualisation/network configuration.

VIRTUALISATION

The VM may reveal that it is running under a hypervisor or another virtual environment.

Today's rule: inspect first → interpret output → record evidence → explain what the evidence means.

WEEK 1 INVENTORY COMMANDS

Run in this order and capture the output.

1. hostnamectl

Identity + OS

2. uname -a

Kernel details

3. ip -br link

Interface/link state

4. ip -br address

Interface addresses

5. ip route

Routing table

6. systemd-detect-virt

Virtualisation type

COMMAND 1: hostnamectl

Inspect host identity and operating-system metadata.

$ hostnamectl

- Static hostname: the persistent name assigned to the host.
- Operating System: identifies the Linux distribution/release when available.
- Kernel: shows the running kernel family/version.
- Architecture: indicates the machine architecture reported by the system.
- Virtualisation information may also appear on some systems.

RECORD

Hostname • OS • kernel • architecture

INTERPRET

Which fields describe the VM itself, and which may reflect the virtual platform?

COMMAND 2: uname -a

Inspect kernel and machine information in one line.

$ uname -a

- Kernel name and release: identifies the running Linux kernel.
- Node name: normally corresponds to the host's network node name.
- Kernel build/version information: indicates how and when the kernel was built.
- Machine architecture: commonly x86_64 on 64-bit Intel/AMD-compatible systems.
- Useful for support, compatibility checks and documenting the environment.

COMPARE

hostnamectl is structured and OS-oriented; uname -a is compact and kernel-oriented.

EVIDENCE

Copy the output exactly, then explain the kernel release and architecture.

COMMAND 3: ip -br link

Inspect network interfaces and their link state.

$ ip -br link

- Interface name: examples may include lo, eth0, ens33 or similar names.
- State: UP or DOWN indicates administrative/link status as reported.
- Loopback (lo): internal interface used by the host to communicate with itself.
- Ethernet/virtual interface: connects the VM to its configured virtual network.
- MAC/link-layer information may be displayed depending on output and version.

QUESTION

Which interface is likely to carry normal traffic outside the VM?

DO NOT ASSUME

Interface names differ across systems and virtualisation platforms.

COMMAND 4: ip -br address

Inspect IP addresses assigned to each interface.

$ ip -br address

- Shows interface name, state and assigned IPv4/IPv6 addresses in compact form.
- 127.0.0.1/8 on lo is the IPv4 loopback address, not the VM's LAN address.
- An address such as 192.168.x.x, 10.x.x.x or 172.16–31.x.x is private IPv4 space.
- The prefix length, for example /24, describes the network prefix.
- The VM's address may be assigned by the virtual network's DHCP service.

RECORD

Interface • state • IPv4 • IPv6 • prefix length

INTERPRET

Which address belongs to the VM and which is only loopback?

COMMAND 5: ip route

Inspect the kernel's IPv4 routing table.

$ ip route

- Default route: the path used when no more-specific route matches the destination.
- via <address>: normally identifies the next-hop gateway for that route.
- dev <interface>: identifies the interface used to send the traffic.
- Connected route: represents a network directly reachable through an interface.
- src <address>: may show the preferred source address for traffic on a route.

RECORD

Default gateway • outgoing interface • connected network

ADMIN VIEW

A correct IP address without a usable route can still leave the host unable to reach remote networks.

COMMAND 6: systemd-detect-virt

Identify whether Linux detects a virtualisation environment.

$ systemd-detect-virt

- Returns a virtualisation/container identifier when a recognised environment is detected.
- Possible results depend on the platform, for example VMware, Oracle/VirtualBox, KVM or a container technology.
- A result indicating none means systemd did not detect supported virtualisation.
- Useful when interpreting hardware, network interfaces and performance in a lab VM.
- Do not treat the detected platform as proof of every configuration detail; record what the command actually reports.

WHY IT MATTERS

The VM's apparent hardware and networking are influenced by the hypervisor configuration.

EVIDENCE

Record the exact returned value.

BUILD THE BASIC SYSTEM & NETWORK INVENTORY

Convert command output into an administrator's record.

Evidence item	Command	What to record
Host identity	hostnamectl	Hostname, OS, kernel, architecture
Kernel detail	uname -a	Kernel release/build, machine architecture
Interfaces	ip -br link	Interface names and state
Addresses	ip -br address	IPv4/IPv6 addresses and prefixes
Routes	ip route	Default gateway, interface, connected route
Virtualisation	systemd-detect-virt	Detected virtualisation type

GOOD ADMINISTRATIVE EVIDENCE

Evidence should allow another person to understand what you observed and what you concluded.

1. CAPTURE

Record the exact command and relevant output. Include enough context to identify the host.

2. INTERPRET

Explain what the output means. Identify the important field, state, address or route.

3. CONNECT

Relate the evidence to an administrative responsibility or infrastructure component.

Weak: screenshot only. Better: command + output + interpretation + administrative meaning.

SAFE LAB PRACTICE

Week 1 commands are primarily inspection commands. Preserve the environment while you learn it.

- Work only in the authorised course VM/lab environment.
- Read the command before pressing Enter; do not paste unknown commands blindly.
- Use least privilege: do not become root merely because you can.
- Do not expose passwords, tokens or sensitive personal data in screenshots.
- Record what you did so that your evidence can be reproduced.
- If output differs from the lecturer's example, investigate before assuming it is wrong.
- Ask before making configuration changes that affect shared resources.

GROUP ACTIVITY: 6–7 GROUPS

Each group selects a small Ghanaian organisation and models what an administrator must manage.

- Choose one small organisation with a clear purpose and manageable scope.
- Identify its main users and user groups.
- Identify the computing systems required to support work.
- Identify network resources required for connectivity.
- Identify essential IT services used or delivered by the organisation.
- Assign likely administrator responsibilities.
- Produce and present a simple infrastructure diagram.

POSSIBLE

Clinic • retail shop • school • law firm • logistics SME • microfinance office •

KEEP IT SMALL

The aim is to understand relationships, not to design an enterprise data centre.

GROUP WORKSHEET 1: USERS

Start with people because infrastructure exists to support work.

- Who uses the organisation's IT resources?
- What different roles exist: management, finance, operations, customer service, technical staff, customers?
- Which users need ordinary access and which need administrative access?
- Are there shared devices, guest users or external service providers?
- What information or services does each user group need?

OUTPUT

List 4–6 user roles and one IT need for each.

ADMIN LINK

Accounts, authentication, permissions, support and access control begin with users.

GROUP WORKSHEET 2: SYSTEMS

Identify the computing hosts and resources that deliver the work.

- Workstations/laptops used by staff.
- Servers or virtual machines hosting applications, files or databases.
- Storage used for operational data and backups.
- Printers or other managed peripherals where relevant.
- Operating systems and key applications at a high level.
- Which systems are critical, and what happens if one fails?

OUTPUT

List the major systems and the service each supports.

ADMIN LINK

Configuration, updates, monitoring, backup, security and troubleshooting.

GROUP WORKSHEET 3: NETWORK RESOURCES

Show how users and systems communicate.

- Internet connection and edge device/router.
- Switching required to connect wired devices.
- Wireless access point(s) where mobility is needed.
- IP addressing and gateway concept at a high level.
- Network segments or separation where clearly justified.
- Links between users, systems and external/cloud services.

OUTPUT

Identify the devices/links and draw the communication path.

ADMIN LINK

Interface state, addressing, routing, availability, documentation and troubleshooting.

GROUP WORKSHEET 4: SERVICES

A service is the useful capability delivered through the infrastructure.

- Business/application service: accounting, records, sales, learning, booking or another core application.
- File and print services where required.
- Authentication or directory service where appropriate.
- DNS and DHCP as supporting network services.
- Email, web or cloud services used by the organisation.
- Backup, logging or monitoring services that support operations.

OUTPUT

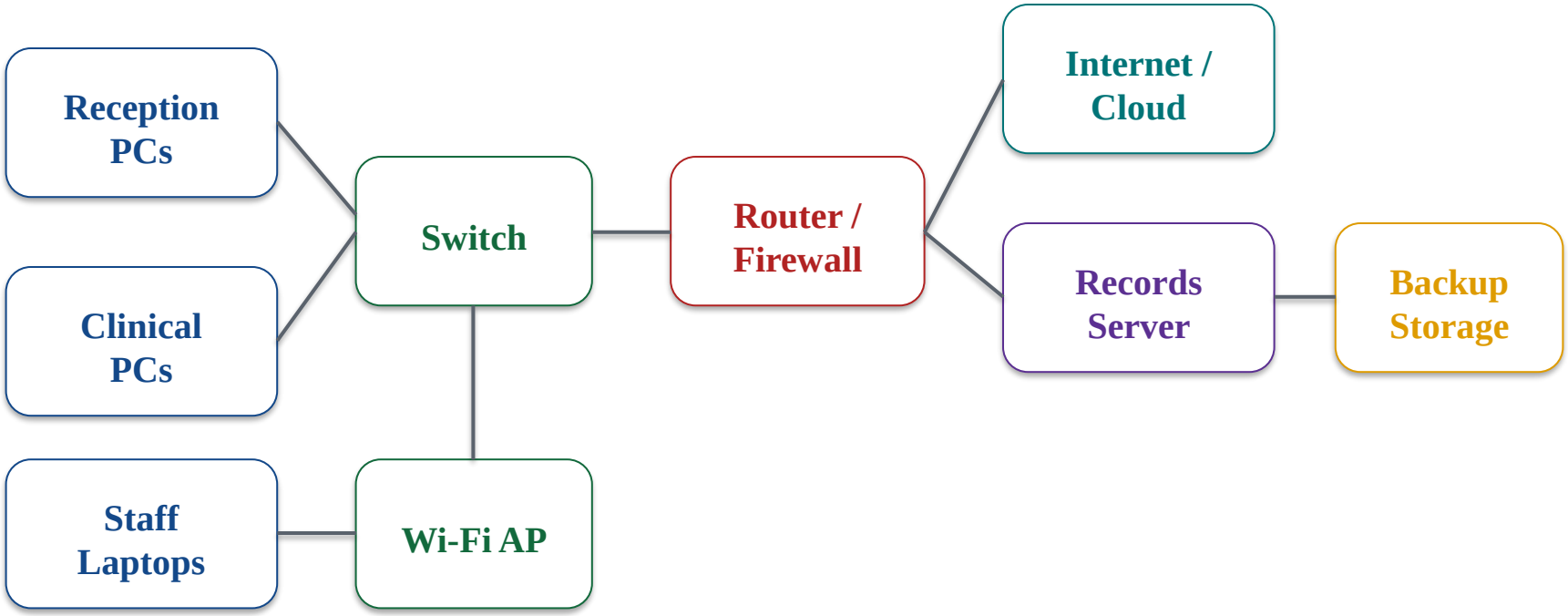
Name the essential services and identify where each runs or is obtained.

ADMIN LINK

Services create dependencies across users, hosts, networks and external providers.

EXAMPLE: SMALL GHANAIAN ORGANISATION

Illustrative infrastructure diagram for “AccraCare Clinic”.



Use labels that show users, systems, network resources and services. Keep the diagram readable and explain the administrator’s responsibility for each major component.

YOUR INFRASTRUCTURE DIAGRAM: MINIMUM STANDARD

A good diagram should answer “what is connected, why, and who manages it?”

- Give the organisation/scenario a clear name.
- Show the main user groups or user locations.
- Show major systems: workstations, servers/VMs, storage as relevant.
- Show network resources: router, switch, AP, Internet/cloud connection as relevant.
- Label essential services and where they are delivered.
- Use arrows/links consistently to show communication paths.
- Mark at least one critical dependency or single point of failure.
- Be prepared to explain administrator responsibility for each major element.

RESPONSIBILITY MAP

For every component, ask four administrative questions.

1. CONFIGURE

What must be set correctly for it to work?

2. MONITOR

What evidence tells us it is healthy or failing?

3. PROTECT

Who should access it and what must be secured?

4. DOCUMENT

What information must another administrator be able to find?

These four questions convert an infrastructure drawing into an administration plan.

GROUP PRESENTATION: 3 MINUTES EACH

Present the environment as an administrator, not as a salesperson.

- 30 sec — Organisation and what it does.
- 45 sec — Main users and systems.
- 45 sec — Network resources and essential services.
- 30 sec — One critical dependency or likely failure point.
- 30 sec — What the administrator must configure, monitor, protect and document.

LECTURER CHECK

Are all four required categories present: users, systems, network resources, services?

PEER QUESTION

What important administrative responsibility has the group overlooked?

WHAT THE INDIVIDUAL PRACTICAL PROVES

Six commands provide a first evidence-based profile of the lab environment.

hostnamectl

Who/what is this host?

uname -a

What kernel and architecture are running?

ip -br link

What network interfaces exist and what is their state?

ip -br address

What addresses are assigned?

ip route

Where will traffic be sent?

systemd-detect-virt

What virtualisation environment is detected?

WHAT THE GROUP ACTIVITY PROVES

Administration begins by understanding the organisation and its dependencies.

- Users define who needs access and support.
- Systems provide computing capacity and host applications/data.
- Network resources connect users, systems and external services.
- Services deliver the capabilities the organisation actually values.
- Documentation makes the environment understandable and supportable.
- Change control protects service stability when the environment evolves.
- Network and systems responsibilities overlap around service availability.

WEEK 1 KEY TAKEAWAYS

The foundation of administration is disciplined responsibility.

1. **Administration manages technology for reliable organisational service.**
2. **Systems and network administration are distinct viewpoints but tightly interdependent.**
3. Administrators are responsible for configuration, operations, security, troubleshooting and evidence.
4. Documentation preserves organisational knowledge; change control reduces avoidable risk.
5. Basic Linux commands can establish a defensible system and network inventory.
6. A simple infrastructure diagram reveals users, systems, network resources, services and dependencies.

WEEK 1 EVIDENCE & NEXT STEP

Submit/retain clear evidence from both the individual and group activities.

INDIVIDUAL EVIDENCE

- hostnamectl output + interpretation
- uname -a output + interpretation
- ip -br link output + interpretation
- ip -br address output + interpretation
- ip route output + interpretation
- systemd-detect-virt output + interpretation

GROUP EVIDENCE

- Organisation/scenario name
- Users and user roles
- Systems
- Network resources
- Essential services
- Simple infrastructure diagram
- Key administrator responsibilities

Reading focus: Chapter 1, “Where to Start.” Next week: computing systems and components from the administrator’s perspective.

WEEK 1 READING & SOURCE ANCHORS

Use the prescribed reading to deepen the administrative perspective.

- Course outline, Week 1: meaning and scope; administrator roles; infrastructure components; documentation; change control; relationship between network and systems work.
- Nemeth, E., Snyder, G., Hein, T. R., & Whaley, B. UNIX and Linux System Administration Handbook, 4th ed., Chapter 1: “Where to Start.”
- Chapter 1 themes used in this deck: account provisioning, hardware, backups, software management, monitoring, troubleshooting, local documentation and security.
- Practical commands and group activity follow the Week 1 course outline supplied for BSIT401.

Important: the uploaded handbook is the 4th edition; the course outline cites the 5th edition. This deck uses the uploaded 4th-edition Chapter 1 as the available textbook source while following the current Week 1 outline.